Subjective Wellbeing and Mental Health: An Exploratory Analysis of a Population Survey in Cali, Colombia

Executive Summary

This study is an exploratory, observational analysis of a population survey from Cali, Colombia, aimed at identifying stable patterns in the relationship between mental health symptoms and subjective wellbeing. It is not designed to estimate the effect of a specific intervention, but rather to provide evidence that may be useful for wellbeing-focused research and interpretation.

The results show a consistent negative association between depression, anxiety, stress, and subjective wellbeing, while demographic factors explain comparatively less variation. Correlations, regression models, prediction checks, and robustness tests all point in the same direction, suggesting that the findings are stable, though they should be interpreted as associations rather than causal effects.

Research Question

This analysis asks whether mental health symptoms are associated with subjective wellbeing in a population survey from Cali, Colombia. In particular, it examines whether depression, anxiety, and stress are linked to lower wellbeing, and whether demographic factors such as age, gender, and socioeconomic status help explain any additional variation.

A second question is whether these relationships remain stable when the model is tested in different ways. To check this, the analysis looks at correlations, regression models, prediction performance, and robustness checks such as outlier analysis and quantile regression.

Data

The dataset comes from a 2022 face-to-face population survey in Cali, Colombia, with 1,281 participants. It includes demographic variables such as birth year, gender, socioeconomic status, race/ethnicity, and parental education. It also includes several wellbeing items and validated mental health measures: the Perceived Stress Scale (PSS-10), the Generalized Anxiety Disorder scale (GAD-7), and the Patient Health Questionnaire (PHQ-9).

The wellbeing measure is especially useful because it captures both life satisfaction and emotional experience. Respondents rated satisfaction with life and also reported how they felt on the previous day across dimensions such as happiness, laughter, enjoyment, worry, depression, stress, and loneliness.

Some variables had missing data, especially parental education. Gender was nearly balanced, and the sample was concentrated in the lower and upper-lower socioeconomic strata. The race/ethnicity variable was also diverse, with a large share of respondents identifying as Black, Afro-descendant, Afro-Colombian, or “none of the above.” These features make the dataset interesting for descriptive analysis, though the main focus here is the relationship between wellbeing and mental health symptoms.

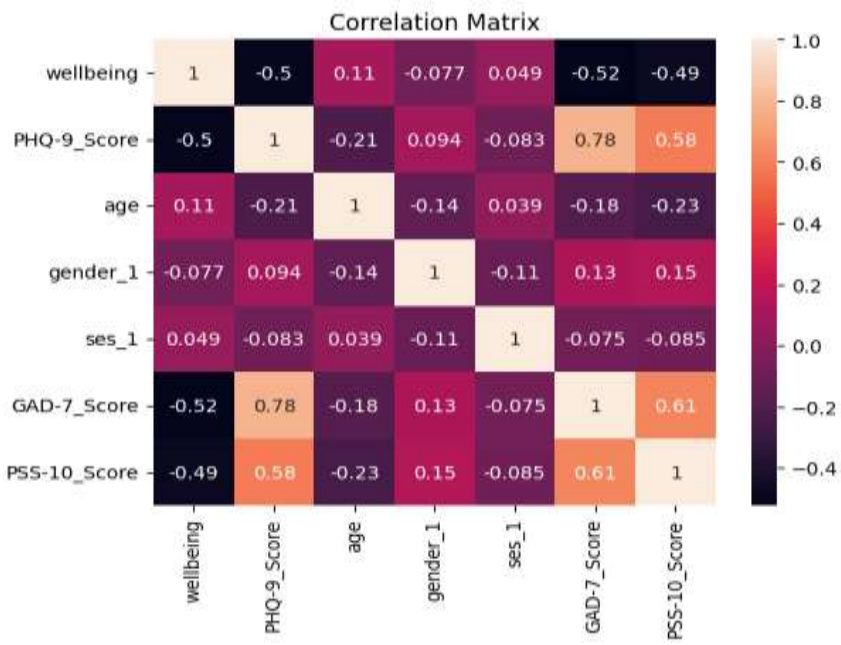
Dataset link: [Mental Health Dataset]

Analysis and Findings

I began with descriptive statistics and missing-data checks to understand the structure of the dataset. After Cleaning the data, I examined **pairwise correlations** between wellbeing and the mental health measures to assess the direction and strength of the main relationships. I created a composite wellbeing score by taking the average of the ten wellbeing items, after reverse-coding the negatively framed questions. I then calculated Pearson correlation coefficients between wellbeing, PHQ-9 depression, GAD-7 anxiety, PSS-10 stress, and the demographic variables.

Correlation Analysis: The correlation matrix showed that wellbeing was moderately negatively correlated with depression (-0.50), anxiety (-0.52), and stress (-0.49). This suggests that higher psychological distress is associated with lower subjective wellbeing. Among the predictors, depression, anxiety, and stress were also strongly correlated with one another, especially depression and anxiety (0.78), which indicates substantial overlap between the mental health measures.

The demographic variables were only weakly correlated with wellbeing, with age showing a small positive association, gender a small negative association, and socioeconomic status a small positive association. Because these correlations were weak, demographic factors do not appear to be major predictors of wellbeing in this sample, especially compared with depression, anxiety, and stress.



Regression analysis: I then estimated a series of OLS regression models to examine how wellbeing was associated with depression, anxiety, stress, and demographic characteristics. In the first model, PHQ-9 depression was a strong negative predictor of wellbeing ($\beta = -0.1586$, $p < 0.001$), and the model explained 25.2% of the variation in wellbeing ($R^2 = 0.252$). This suggests a clear and statistically robust association between higher depression scores and lower wellbeing.

I next added age, gender, and socioeconomic status to the model. The effect of depression remained virtually unchanged ($\beta = -0.1578$, $p < 0.001$), while age, gender, and socioeconomic status were not statistically significant predictors of wellbeing. The model fit improved only marginally ($R^2 = 0.253$), indicating that these demographic variables added little explanatory power.

In the full model, I included PHQ-9 depression, GAD-7 anxiety, PSS-10 stress, age, gender, and socioeconomic status together. All three mental health variables remained significant negative predictors of wellbeing: depression ($\beta = -0.0572$, $p < 0.001$), anxiety ($\beta = -0.0841$, $p < 0.001$), and stress ($\beta = -0.0654$, $p < 0.001$). Anxiety had the strongest negative association, followed by

stress and then depression. The full model explained 33.1% of the variation in wellbeing ($R^2 = 0.331$), showing a meaningful improvement over the simpler models.

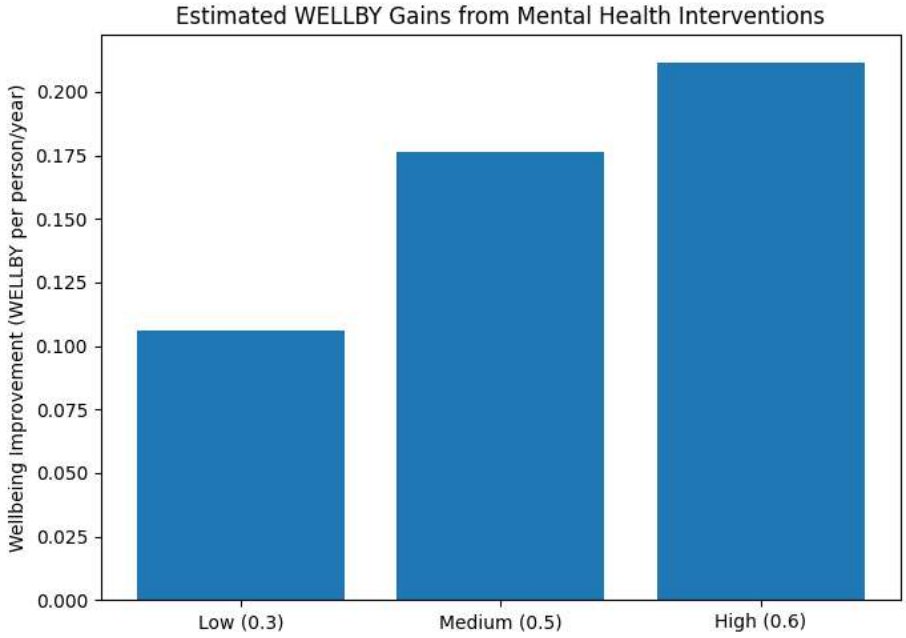
Overall, the regression results suggest that mental health symptoms are much stronger predictors of wellbeing than demographic characteristics in this sample. The pattern is consistent with the correlation analysis and remains stable across model specifications, which increases confidence in the main conclusion.

I then estimated an OLS regression model to see whether age, gender, and socioeconomic status were associated with perceived stress. The model explained a small share of the variation in stress ($R^2 = 0.072$), suggesting limited overall explanatory power. Age was negatively associated with stress ($\beta = -0.0934, p < 0.001$), socioeconomic status was also negatively associated with stress ($\beta = -0.3701, p = 0.018$), and gender was positively associated with stress ($\beta = 1.5630, p < 0.001$). Overall, the results suggest that demographic factors do matter for stress, but only modestly.

Outcome	**Key predictors**	**Main finding**	**R-squared**
Wellbeing	PHQ-9_Score	Higher depression was linked to lower wellbeing.	0.252
Wellbeing	PHQ-9_Score, age, gender, SES	Depression still mattered, but age, gender, and SES did not add much.	0.253
Wellbeing	PHQ-9_Score, GAD-7_Score, PSS-10_Score, age, gender, SES	Depression, anxiety, and stress were all linked to lower wellbeing, with anxiety showing the strongest link.	0.331
Stress	age, gender, SES	Older participants and those with higher SES tended to report less stress, while one gender group reported more stress.	0.072

I also assessed the model's predictive performance using a train-test split. The correlation between actual and predicted wellbeing was 0.55, indicating moderate predictive ability. This suggests that the model can capture some of the main relationships in the data, but additional factors would be needed to predict wellbeing more accurately.

Wellbeing estimation: I also estimated the potential wellbeing gains from mental health improvements by translating intervention effects into WELLBYs. Using the regression coefficient from the wellbeing model and the standard deviation of PHQ-9 scores, I estimated the expected wellbeing gains for three illustrative effect sizes. The low-effect scenario corresponds to about 0.105 WELLBYs per person per year, the medium-effect scenario to about 0.177 WELLBYs, and the high-effect scenario to about 0.212 WELLBYs. The pattern suggests that larger mental health improvements could produce meaningful gains in wellbeing, although these estimates should be treated as illustrative rather than exact intervention effects.



The pattern is straightforward: higher intervention effectiveness is associated with larger wellbeing gains. The increase is not perfectly linear, because the jump from low to medium effectiveness is larger than the jump from medium to high, which suggests diminishing marginal gains at higher levels of effectiveness. In practical terms, this means that improving weaker programs to a moderate level may deliver the biggest increase in wellbeing, while further improvements to already strong programs still matter but add less additional benefit.

Outlier Analysis: I also checked the data for outliers to see whether a small number of unusual observations might be affecting the results. This step is important because extreme values can sometimes distort regression estimates and make the model look stronger or weaker than it really is. After reviewing the outliers, I tested the model again and found that the main findings did not change much. This suggests that the overall relationship between mental health and wellbeing was not being driven by a few extreme cases, which makes the results more reliable.

Quantile regression: After checking the model for outliers, I used quantile regression as a robustness test to see whether the main results held across different parts of the wellbeing distribution. This step helped confirm that the findings were not being driven only by a few extreme observations. Overall, the quantile regression results supported the main pattern from the OLS models, suggesting that the relationship between mental health and wellbeing was reasonably stable even when the model was tested more carefully.

Interpretation

The main pattern in this analysis is that mental health symptoms are much more closely linked to subjective wellbeing than demographic factors are. In practical terms, this means that depression, anxiety, and stress appear to matter more for how people feel about their lives than age, gender, or socioeconomic status alone. That is an important result because it suggests that improving mental health may be one of the most direct ways to increase wellbeing.

The regression results reinforce this point. Even after accounting for demographic characteristics, the mental health measures remained strongly associated with lower wellbeing. This suggests that the connection between psychological distress and subjective wellbeing is not just a byproduct of background differences in the sample. Instead, it appears to be a robust relationship that holds across several model specifications.

The prediction results show that the model could capture the general pattern in the data, but it did not explain everything. That is not surprising in a wellbeing study, since wellbeing is influenced by many factors that were not included here. The moderate predictive performance suggests that the model is useful for identifying broad relationships, but not enough to fully predict individual wellbeing scores.

The WELLBY estimates translate the statistical results into a more policy-relevant format. They suggest that stronger mental health improvements can lead to larger wellbeing gains, which supports the idea that mental health interventions may produce meaningful benefits beyond symptom reduction alone. The pattern is especially important because it shows that moving from weaker to moderate intervention quality may deliver larger gains than fine-tuning already strong programs.

The robustness checks add confidence to the analysis. The main findings did not change much after examining outliers and using quantile regression, which suggests that the results are not being driven by a few unusual observations. Taken together, this makes the overall story more credible: mental health is strongly tied to wellbeing, and the relationship appears stable across different checks.

Limitations

This analysis has several limitations. First, it is based on a population survey from Cali, Colombia, so the results may not generalize beyond this setting. Second, it uses cross-sectional survey data, so it cannot show causal relationships. Third, the measures of depression, anxiety, and stress are closely related to one another, which makes it difficult to separate their individual effects very precisely. The wellbeing estimate is also based on an illustrative conversion into WELLBYs, so it should be treated as an approximation rather than a precise intervention effect.

The model also leaves a meaningful amount of wellbeing unexplained, which means other factors not included in the analysis are likely important. In addition, the results rely on self-reported data, which can be affected by reporting bias and measurement error. Although the robustness checks.

Conclusion

Overall, the analysis of a population survey from Cali, Colombia shows a consistent negative relationship between mental health symptoms and subjective wellbeing. Depression, anxiety, and stress were all linked to lower wellbeing, while demographic factors were much less important. The prediction results, WELLBY estimates, and robustness checks all pointed in the same direction, which strengthens confidence in the main message.

GitHub repository: https://github.com/najiyak23/Mental-Health-and-Subjective-Wellbeing-Analysis/blob/main/Mental_Health_Analysis.ipynb

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